Scientific Justification *Limited to 1 page of text plus 2 pages for figures and references.*

Executive Summary: We propose a high-cadence DECam imaging program to measure optical variability in a sample of dwarf galaxy active galactic nuclei (AGN) hosting intermediate-mass black holes (IMBHs). These systems provide critical constraints on black hole seed formation and the occupation fraction of black holes in low-mass galaxies, yet their variability properties remain largely unexplored. Using a sample of 229 spectroscopically identified dwarf AGN candidates from SDSS, we select a subset of 11 dwarf galaxies within 5 DECam fields. We propose observations in g and r bands with a cadence of 7 exposures per week over a 10 week baseline, enabling sensitivity to variability on days–weeks timescales. Exposure times are chosen based on DECam Exposure Time Calculator (ETC) estimates to achieve high signal-to-noise, resulting in ∼10s (r) and ∼15s (g) exposures (+ 30s overheads). The total requested time is ∼8.5 hours. This program will (1) measure the variability fraction of IMBH AGNs, (2) characterize variability amplitudes as a function of black hole mass and accretion rate, and (3) test the ability of multi-band variability to distinguish AGNs from transient contaminants. The results will provide the first systematic constraints on short-timescale variability in low-mass AGNs and establish variability as a robust tool for confirming IMBH candidates.

Motivation: Active galactic nuclei (AGN) in dwarf galaxies provide a crucial probe of black hole seed formation and the occupation fraction of black holes in low-mass systems (Reines & Comastri 2016; Greene et al. 2020). Intermediate-mass black holes (IMBHs; 10^4–$10^6\,M_\odot$) are particularly important in this context, yet their demographics and accretion properties remain poorly constrained.

Spectroscopic surveys have identified candidate IMBH AGNs through broad emission lines (Greene & Ho 2007; Baldassare et al. 2015, 2017), as well as through emission-line diagnostics in dwarf galaxies (Reines et al. 2013). However, these methods can suffer from contamination by star formation or transient phenomena, and independent confirmation of AGN activity is often required.

Optical variability provides a powerful and independent diagnostic of accretion onto black holes. Time-domain surveys, such as the Subaru Hyper Suprime-Cam (HSC), have demonstrated that variability selection can robustly identify AGNs and characterize their stochastic behavior (e.g., Akiyama et al. 2018; Tachibana et al. 2020, Burke et al. 2021). However, these studies focus primarily on variability-selected samples rather than spectroscopically confirmed IMBH AGNs. As a result, the variability fraction and scaling relations for low-mass AGNs are particularly on short (days to-weeks) timescales where low-mass systems are expected to vary more rapidly.

We propose a high-cadence, multi-epoch imaging program to measure optical variability in a sample of dwarf galaxy AGNs hosting IMBHs. Our program addresses three key questions: (1) what fraction of spectroscopically identified IMBH AGNs exhibit variability on days–weeks timescales, (2) how variability amplitude scales with black hole mass and accretion rate, and (3) whether multi-band variability can distinguish AGNs from contaminating transients such as supernovae.

The proposed observations will provide systematic constraints on short-timescale variability in IMBH AGNs and establish variability as a robust diagnostic for confirming low-mass AGN candidates.

Science Goals:

1. What fraction of spectroscopically-identified broad-line IMBH AGNs show optical variability on days-to-months timescales?

2. How do variability amplitude/timescale scale with BH Mass/Luminosity at low BH masses?

3. Can photometric variability surveys distinguish broad H-alpha AGNs from interlopers (i.e SNe)?

Target	RA	Dec	z	g	r	BH Mass	Luminosity
	°	°		mag	mag	$\log(M_\odot)$	$\log(L_{bol})$
J073106.86+392644.6	112.7786	39.4457	0.048	15.89	15.20	6.100	-0.800
J073505.65+423545.6	113.7735	42.5960	0.086	18.34	17.75	6.200	-0.400
J074423.44+243046.3	116.0977	24.5129	0.117	18.01	17.19	6.200	-1.100
J074831.91+464455.4	117.1330	46.7487	0.088	19.67	19.19	6.300	-0.700
J074836.80+182154.2	117.1533	18.3651	0.071	18.10	17.11	6.100	-1.400
J082106.95+404836.5	125.2790	40.8101	0.124	19.20	18.76	6.100	-0.300
J082325.91+065106.4	125.8580	6.8518	0.072	17.02	16.43	5.800	-1.300
J102703.90+485023.9	156.7662	48.8400	0.148	18.32	17.43	6.300	-0.900
J102957.76+431430.8	157.4907	43.2419	0.085	17.53	16.93	6.000	-0.500
J105424.29+454130.5	163.6012	45.6918	0.064	17.28	16.67	5.400	-0.800
J105755.66+482502.0	164.4819	48.4172	0.073	18.19	17.68	5.800	-0.800

References

[1] Akiyama, M., et al. 2018, PASJ, 70, S34
[2] Baldassare, V. F., et al. 2015, ApJ, 809, L14
[3] Baldassare, V. F., et al. 2017, ApJ, 836, 20
[4] Burke, C. J., et al. 2021, ApJ, 919, 142
[5] Greene, J. E., & Ho, L. C. 2007, ApJ, 670, 92
[6] Greene et al. 2020, ARA&A, 58, 257
[7] Reines, A. E., Greene, J. E., & Geha, M. 2013, ApJ, 775, 116
[8] Reines, A. E., & Comastri, A. 2016, PASA, 33, e054
[9] Tachibana, Y., et al. 2020, ApJ, 903, 54

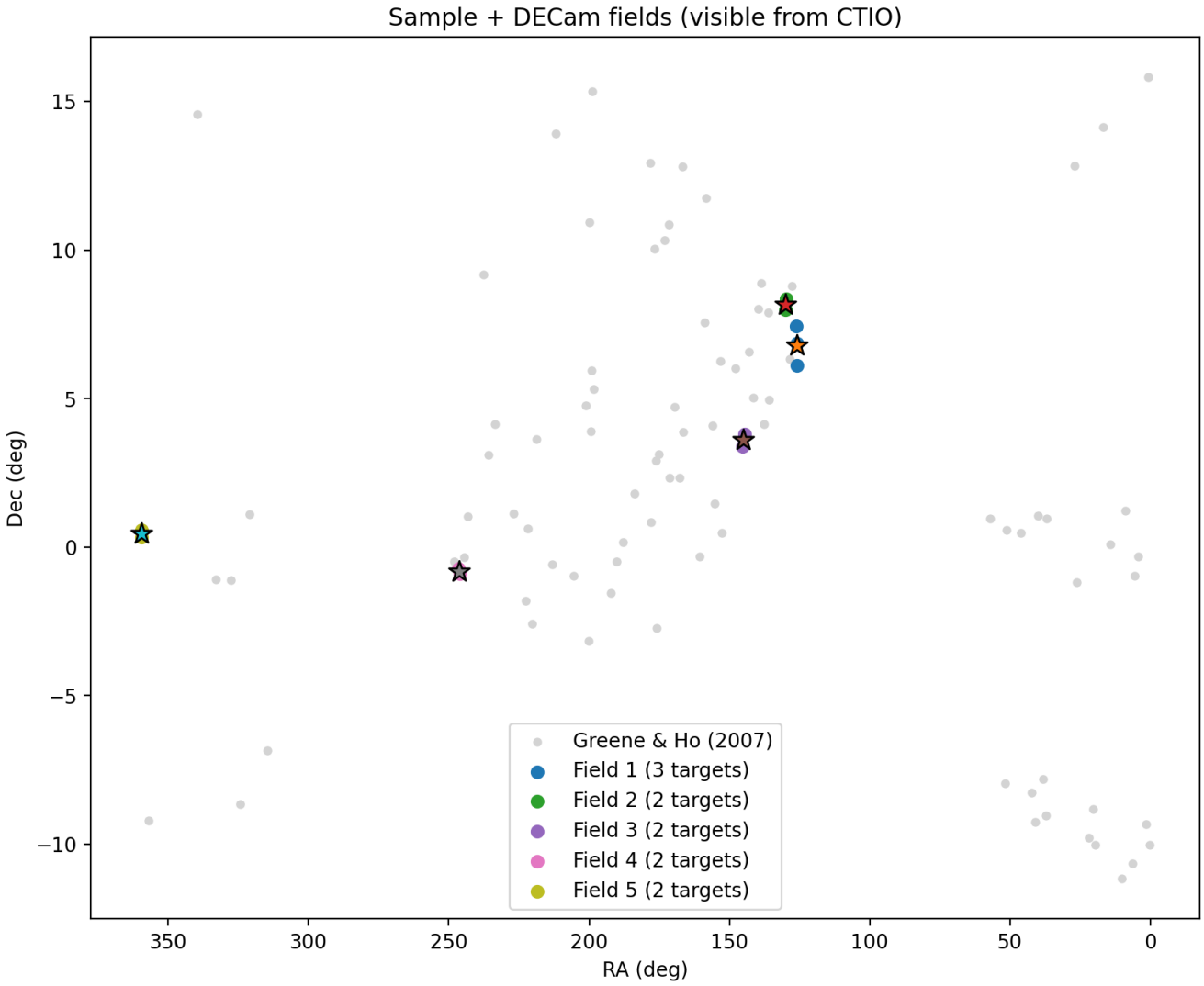

Figure 1: Will make this plot nicer

Experimental Design *Limited to 1 page of text*

We propose a pilot time-domain monitoring program for 11 CTIO-visible dwarf galaxy AGN candidates distributed across 5 DECam fields. The sample was selected from a larger parent catalog using redshift, brightness, and declination cuts, then restricted to DECam footprints that cover multiple objects within one exposure. This yields a compact sample that is well suited to a focused variability experiment while still spanning the low-mass AGN regime.

The goal of the experiment is to test whether spectroscopically identified low-mass AGNs exhibit measurable optical variability on day-to-week timescales. We will obtain multi-epoch photometry in both g and r bands, using the r band as the primary variability tracer and the g-band to provide color information that can help distinguish AGN-like variability from transient contamination. Exposure times are calculated to produce a signal-to-noise ratio ≥ 40.

We adopt a daily cadence (7 times per week) on a $\sim$10-week baseline. This cadence is sufficient to sample short-time-scale fluctuations while also capturing repeat variability behavior across the observing window. The program is designed as a pilot study: even with only 11 targets, repeated observations over multiple epochs will allow us to measure variability amplitudes, compare behavior across bands, and determine whether these low-mass nuclei are consistent with stochastic AGN variability or with transient-like evolution. Additionally, with DECam's large FOV we can search for other objects that exhibit short timescale variability (day-to-week) in these 5 field exposures.

Technical Description *Limited to 1 page of text*

The observations will be carried out with DECam on the CTIO Blanco 4-m telescope. DECam is well suited to this program because its wide field of view allows multiple targets to be monitored simultaneously within each pointing, making it efficient for time-domain work on a sparse but sky-localized dwarf AGN sample. The final program requires 5 DECam fields to cover the 11 selected targets.

Observations will be obtained in both the g and r bands. Based on DECam ETC calculations, we adopt exposure times of 15s in g and 10s in r for the representative targets in the sample. These exposure times will provide adequate signal-to-noise ratio (≥ 40) for variability measurements while keeping the observing sequence efficient. We will use the ETC values derived for our dimmest object in either band, which in our case is a g band magnitude of $\sim$20.

We request typical good-image-quality conditions, with seeing of approximately $1''$ and airmass less than 1.5. These conditions are appropriate for precision differential photometry and ensure stable image quality across the multi-epoch campaign. Because AGN variabilty relies on repeated monitoring, we will prioritize the consistency of observing cadence over extremely deep individual exposures.

DECam overheads must be included separately from the exposure time estimations from the ETC. For each exposure, we adopt an overhead of approximately 30 seconds, which includes CCD readout and the usual filter/telescope transitions. Thus, the effective time per exposure is approximately 40s in r and 45s in g. The total program time is therefore determined by the number of fields, filters, and visits across the full baseline. For a 5-field program observed 7 times per week (daily) over $\sim$10 weeks in both filters. Therefore, the total time we request is $\sim$8.5 hours over 10 weeks.

The observations will yield multi-epoch light curves in both filters for all 11 targets. The resulting exposures from our program will be used to measure flux variability, assess band-to-band behavior, and evaluate whether the targets show AGN-like stochastic variability or transient-like changes. The combination of wide-field imaging, short exposures, and repeated visits makes DECam an efficient and technically well-matched instrument for this experiment.

Things to add

Band 3 Plan *Limited to 1/2 page of text*

This program is not suitable for band 3.

Classical Backup Program *Limited to 1/2 page of text*

This is not a classical request.

Justify Target Duplications

Need to figure this out

ITC Examples

DECam EXPOSURE TIME CALCULATOR (ETC) VER 7B

History

December 10 2008	Written by Daren DePoy
January 26 2012	QE, filter responses updated, prettified by Alistair Walker
Apr 8 2013	Table of preliminary values of sky brightness as a function of lunar phase provided
April 13 2013	Added in Read noise, split exposures, correct u band throughput
September 30 2013	Ver 4: Corrected effective mirror area from 10.6 m2 to 9.7m2 and set vignetting to
March 4 2014	Added single cell lunar phase selection and sky brightness interpolation (RCS)
September 1 2014	Ver 5: Corrected the aperture used to follow the true profile (sech^2) rather than a
March 1 2015	Ver 6: Revised QE from the definitive paper, and real-world relectivity of M1
March 8 2016	Ver 7. Corrected error in signal/noise due to read noise, revised the value of the R(

Instructions

(i) Change exposure time and/or object magnitude, high-lighted with YELLOW background

(ii) Choose your lunar phase in days (0 = New Moon, 14 = Full Moon)

(iii) Read on last line the S/N of that brightness in that exposure time, highlighted with BLUE background

Moon phase = 14 (number between 0 and 14, inclusive)

filter	DECam u	DECam g	DECam r	DECam i
exposure time (sec)	**20**	**15**	**10**	**20**
Sky brightness mag	**17.7**	**19.4**	**19.7**	**19.4**
bandpass	50	147	141	147
bandpass	107068735000000	201414494177331	104965177541270	73942347668952
cwl	375	473.5	638.5	775.5
hc/lambda	5.30E-19	4.20E-19	3.11E-19	2.56E-19
0 mag	7.34E+09	1.74E+10	1.23E+10	1.05E+10
0 mag	1.47E+11	1.19E+11	8.69E+10	7.13E+10
sky surface brightness	6.47E-15	8.65E-16	3.56E-16	3.17E-16
sky surface brightness	2.75E-04	3.68E-05	1.52E-05	1.35E-05
sky surface brightness	2.75E-08	3.68E-09	1.52E-09	1.35E-09
telescope area	9.70	9.70	9.70	9.70
0 mag	1.42E+12	2.54E+12	1.19E+12	2.03E+12
CCD qe	0.25	0.59	0.75	0.85
filter	0.95	0.9	0.9	0.9
corrector optics	0.75	0.86	0.86	0.86
telescope primary	0.89	0.89	0.88	0.87
vignetting	1	1	1	1
atmosphere	0.7	1.5	1.5	0.9
total throughput	0.11	0.61	0.77	0.52
Signal from 0 mag source	1.58E+11	1.55E+12	9.11E+11	1.05E+12
Signal from 20 mag source	1580	15463	9105	10475
Signal from specified source	1580	15463	9105	10475
seeing	1.31	1.06	1.00	0.96
aperture (=2.04*seeing)	5.63	3.68	3.27	3.02

sky signal	7.40E+04	9.90E+04	3.92E+04	5.50E+04
Readout noise e-/pix	7.0	7.0	7.0	7.0
RON squared per aperture	3957.4	2589.8	2297.9	2126.0
Number of reads	1	1	1	1

brightness (AB mag)	**"----------S/N in time entered in line 24----------**			
20	5.6	45.2	40.5	40.3

PRELIMINARY SKY BRIGHTNESS TABLE FOR DECam - caveat > 50 deg

MOON days from NEW	**u**	**g**	**r**	**i**
0	22.8	22.1	21.1	20.1
3	22.3	21.8	20.9	20.1
7	20.7	21.2	20.6	19.9
10	19.2	20.4	20.2	19.7
14	17.7	19.4	19.7	19.4

mags(AB) per arcsec^2